Randomized and low-rank approximation

Nonadaptive queries for finite-family matrix approximation

Difficulty: challenging **Importance:** interesting to the community

Status: open; checked 2026-09-08
Area: nonadaptive matrix sketching

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab \min(a, b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^{\mathsf{T}}v$ for one chosen real vector v; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm's randomness.

Problem statement

Let $n \geq 1$, let $\mathcal{F} \subset \mathbb{R}^{n\times n}$ be explicitly given with $M = |\mathcal{F}| \geq 2$, and let $A \in \mathbb{R}^{n\times n}$ be accessible only through matrix–vector queries. Set $t = \log(2M)$.

Question

Do absolute constants $C > 0$, integer $b \geq 0$, and a uniform randomized algorithm exist that, for every $0 < \varepsilon < 1/2$, choose all query vectors and all choices between A and A^{T} before receiving any oracle answers, make at most

$$C\sqrt{t}\,\varepsilon^{-2}[1 + \log(2 + t) + \log(1/\varepsilon)]^{b}$$

queries, and return $B \in \mathcal{F}$ such that

$$\Pr\Big[\|A - B\|_F \leq (3 + \varepsilon) \min_{D\in\mathcal{F}} \|A - D\|_F\Big] \geq 0.99?$$

Query choices may depend on $\mathcal{F}$, ε and randomness. Only oracle calls are charged; candidate processing and processing of the answers are unrestricted.

References

Amsel et al., *Query Efficient Structured Matrix Learning*, §5, adaptivity question. Christopher Musco, *Structured Matrix Learning from Matrix-Vector Products*, ICERM, February 4, 2026, numbered slide 23.

Status evidence

Both sources explicitly ask whether adaptivity is necessary; the paper's August 21, 2026 revision retains the question. Searches for "finite matrix approximation adaptivity 2026" and the paper identifier with "non-adaptive" found no resolution. The abstract's linked finite-family improvement, Theorem 1.2, uses two rounds after receiving a constant-factor warm start: its left query vectors depend on the first round's answers. It does not supply the required nonadaptive algorithm.